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PLASMA POTENTIAL ADJUSTMENT CIRCUIT

Abstract

A method of adjusting sheath potential of a plasma at a substrate includes applying pulsed plasma source power to a plasma electrode in contact with the plasma while applying pulsed bias power to the substrate, measuring electric potential produced by the pulsed plasma source power and the pulsed bias power to obtain time-resolved electric potential information, and adjusting the electric potential profile at the plasma electrode based on the time-resolved electric potential information. Adjusting the electric potential profile adjusts the sheath potential of the plasma at the substrate. The method may be performed using a plasma system including the plasma electrode and a potential adjustment circuit coupled between the plasma electrode and at least one pulsed DC power supply. The potential adjustment circuit may be configured to adjust the electric potential profile at the plasma electrode.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates generally to plasma systems, and, in particular embodiments, to systems and methods for adjusting sheath potential of a plasma at a substrate using a potential adjustment circuit.

BACKGROUND

[0002] Device formation within microelectronic workpieces can involve a series of manufacturing techniques including formation, patterning, and removal of a number of layers of material on a substrate. In order to achieve the physical and electrical specifications of current and next generation semiconductor devices, processing flows enabling reduction of feature size while maintaining structural integrity is desirable for various patterning processes. As device structures densify and develop vertically, the desire for precision material processing becomes more compelling.

[0003] Plasma processes are frequently used during device formation, such as plasma-driven and plasma-enhanced processes including etching processes, deposition processes, cleaning processes, surface modification processes, and others. Electrical power may be pulsed to various structures to generate plasma and/or to influence plasma species in desirable ways for the furtherance of a given plasma process. For example, source power may be applied to one or more structures in a plasma system to generate a plasma. Bias power is often applied to structures electrically coupled to a substrate in order to direct charges species (such as positive ions) towards to substrate to process the substrate.

[0004] When ions are used to process the substrate, it is desirable to deliver the ions to the substrate with certain properties that are favorable for achieving the desired result of the plasma process. For example, the ions may arrive at the substrate with a certain ion energy distribution (IED) and ion angle distribution (IAD), and in a certain quantity (ion flux). Each of these properties may vary both temporally (time-dependence) and spatially (e.g., across different regions of the substrate, such as center versus edge, and so on).

[0005] Ion energy is one property of the ions at the substrate surface that is important to plasma processes (e.g., plasma etching processes, such as high-aspect ratio etching processes like high-aspect ratio contact (HARC) etches, etc.). Many conventional methods of applying power to structures in the plasma system result in an ion energy distribution function (IEDF) that includes an undesirably broad ion energy peak. Narrowing of the IED around at a target ion energy has various benefits, such as improving process uniformity, repeatability, reliability, yield, and others. Moreover, a narrower IED may enable critical dimensions (CDs) and structures that were previously impractical with broader IED plasma processes.

[0006] Therefore, improved systems and methods capable of narrowing the IED of ions at a substrate during a plasma process are desirable.

SUMMARY

[0007] In accordance with an embodiment of the invention, a method of adjusting sheath potential of a plasma at a substrate includes applying pulsed plasma source power to a plasma electrode in contact with the plasma while applying pulsed bias power to the substrate, measuring electric potential produced by the pulsed plasma source power and the pulsed bias power to obtain time-resolved electric potential information, and adjusting the electric potential profile at the plasma electrode based on the time-resolved electric potential information. Adjusting the electric potential profile adjusts the sheath potential of the plasma at the substrate.

[0008] In accordance with another embodiment of the invention, a plasma system includes a plasma electrode configured to be in contact with a plasma, a substrate holder configured to support

a substrate in contact with the plasma, at least one pulsed direct current (DC) power supply configured to deliver pulsed plasma power to the plasma electrode and pulsed bias power to the substrate holder, and a potential adjustment circuit coupled between the plasma electrode and the at least one pulsed DC power supply. The potential adjustment circuit is configured to adjust the electric potential profile at the plasma electrode.

[0009] In accordance with still another embodiment of the invention, a plasma system includes a plasma electrode configured to be in contact with a plasma, a substrate holder configured to support a substrate in contact with the plasma, at least one pulsed power supply, a potential adjustment circuit coupled between the plasma electrode and the at least one pulsed power supply, at least one sensor, and a controller operatively coupled to the potential adjustment circuit and the at least one pulsed power supply. The controller includes a processor and a non-transitory computer-readable medium storing a program including instructions that, when executed by the processor, perform a method that includes applying pulsed plasma power to the plasma electrode while applying pulsed bias power to the substrate using the at least one pulsed power supply, measuring electric potential produced by the pulsed plasma power and the pulsed bias power to obtain time-resolved electric potential information using the at least one sensor, and adjusting an electric potential profile at the plasma electrode based on the time-resolved electric potential information using the potential adjustment circuit. Adjusting the electric potential profile adjusts the sheath potential of the plasma at the substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0011] FIG. 1 illustrates an example plasma system including a plasma electrode configured to be in contact with a plasma and a potential adjustment circuit coupled between the plasma electrode and a pulsed power supply in accordance with embodiments of the invention;

[0012] FIGS. 2A and 2B illustrate qualitative graphs of electric potential and IEDF for an example plasma system where FIG. 2A shows a broad IEDF resulting from an unadjusted plasma potential profile and FIG. 2B shows a narrow IEDF resulting from an adjusted plasma potential profile in accordance with embodiments of the invention;

[0013] FIG. 3 illustrates an example plasma system including a plasma electrode, a potential adjustment circuit, one or more sensors configured to obtain time-resolved electric potential information received by a controller configured to control the potential adjustment circuit in accordance with embodiments of the invention;

[0014] FIG. 4 illustrates an example capacitively coupled plasma (CCP) plasma system that includes a plasma electrode, a potential adjustment circuit, one or more sensors, and a controller in accordance with embodiments of the invention;

[0015] FIG. 5 illustrates an example inductively coupled plasma (ICP) plasma system that includes a source power coupling element, a plasma electrode, a potential adjustment circuit, one or more sensors, and a controller in accordance with embodiments of the invention; and

[0016] FIG. 6 illustrates an example method of adjusting sheath potential of a plasma at a substrate in accordance with embodiments of the invention.

[0017] Corresponding numerals and symbols in the different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the embodiments and are not necessarily drawn to scale. The edges of features drawn in the figures do not necessarily indicate the termination of the extent of the feature.

[0018] The making and using of various embodiments are discussed in detail below. It should be appreciated, however, that the various embodiments described herein are applicable in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use various embodiments, and should not be construed in a limited scope. Unless specified otherwise, the expressions “around”, “approximately”, and “substantially” signify within 10%, and preferably within 5% of the given value or, such as in the case of substantially zero, less than 10% and preferably less than 5% of a comparable quantity.

[0019] Controllable ion energy with narrow energy spread (i.e., a narrow IED about a target ion energy) is highly desired to improve various plasma processes, such as those used during device formation. For example, future nodes may require plasma etching processes that have much tighter control and improved capabilities, such as for the formation of 3-dimensional devices. Some plasma etching process that may benefit from a narrower IED include etching process for gate formation and high-aspect ratio etching for contacts (e.g., HARC etching processes). Ion energy that is delivered with increased control over the energy and a narrower IED is expected to deliver improved etching performance (e.g., less bowing, less aspect ratio-dependent etching (ARDE), etc.), which may be especially important for high aspect ratio etching.

[0020] Many plasma systems deliver pulses of power to the various structures to perform the plasma processes (referred to as pulsed plasma systems herein). Any of the delivered power in the pulsed plasma system may be pulsed, such as the source power, the bias power, and others. One type of pulsed plasma system uses both pulsed source power and pulsed bias power, and the two may be delivered synchronously, antisynchronously, or with arbitrary relative timing. Generating and/or processing a substrate with pulsed power naturally involves transients because the applied power is designed to influence plasma species and the power is being repeatedly switched on and off, sometimes at high pulse frequencies (e.g., in the hundreds of kilohertz, such as about 400 kHz or even higher). Additionally, materials of the substrate (e.g., at the surface) often includes dielectrics, which cause the electric potential within the system to be time-dependent during each pulse.

[0021] Specifically, in pulsed plasma systems, each negative pulse of the bias power is an ion charging phase where positive charge collects at the surface of the substrate. The collected positive charge due to the surface charging effect causes the electric potential at the substrate to have a positive slope during the ion charging phase. Meanwhile, the plasma potential during the ion charging phase is substantially flat or may have a negative slope. Because the plasma sheath potential (specifically between the plasma and the substrate) is equal to the plasma potential minus the substrate potential, the resulting sheath potential has a positive slope (i.e., is changing over time during the negative pulses of the bias power when ions are being accelerated by the plasma sheath towards the substrate). This broadens the IED of the ions because they experience a changing accelerating voltage throughout the pulse. Further, a deeper slope of the substrate potential is often associated with increased broadening of the IED (such as when the plasma potential during the ion charging phase has a negative slope, when the ion flux is higher so charging occurs faster, etc.).

[0022] Conversely, ions accelerated by a constant plasma sheath potential (i.e. a flat slope) have a desirably narrow IED (although some ions are in the plasma sheath at the start and the end of the negative pulses, which causes some disparity in the acceleration time). Therefore, it is desirable to apply power in such a way as to produce a plasma sheath potential with a flat slope during the ion charging phase. However, because of at least the surface charging effects discussed above, merely conventionally applying pulsed power (e.g., including source power and bias power) whether using radio frequency (RF) bias power, DC bias power, dual bias power, etc.) cannot create a constant substrate-plasma plasma sheath potential (regardless of whether the pulsed power is applied to a top electrode, a bottom electrode or both electrodes).

[0023] In accordance with various embodiments herein described, the invention proposes systems

and methods for modulating the plasma potential to shape (e.g., narrow) the IED of ions at the substrate. In particular, the electric potential of the plasma system may be monitored and the electric potential profile of pulsed power applied to a plasma electrode (i.e., the shape of the power applied to the plasma electrode) may be adjusted accordingly to adjust the sheath potential of the plasma at the substrate. The IED of the ions determined at least in part by the sheath potential. Therefore, control over the sheath potential may advantageously facilitate control over the IDE.

[0024] In various embodiments, a plasma system includes a plasma electrode configured to be in contact with a plasma, at least one pulsed power supply configured to deliver pulsed plasma power to the plasma electrode and pulsed bias power to a substrate (e.g., one or more pulsed direct current (DC) power supply, but pulsed alternating current (AC) power supplies may also be used), and a potential adjustment circuit (e.g., including various passive elements such capacitors, inductors, and resistors) coupled between the plasma electrode and the at least one pulsed power supply. The potential adjustment circuit is configured to adjust an electric potential profile of the pulsed plasma power at the plasma electrode. For example, the pulsed bias power may be delivered to a substrate holder configured to support the substrate so that the substrate is in contact with the plasma.

[0025] One or more sensors may also be included and configured to measure electrical potential produced by the pulsed plasma power and the pulsed bias power to obtain time-resolved electric potential information. A controller may be operatively coupled to the one or more sensors and the potential adjustment circuit. The controller may be configured to adjust the electric potential profile of the pulsed plasma power based on the time-resolved electric potential information to adjust the sheath potential of the plasma at the substrate.

[0026] Conventional systems and methods used for plasma processes that are seeking to narrow the IED of ions delivered to the substrate being processed primarily focus on applying different (and specially tailored) waveforms to control the substrate potential, and thereby control ion energy to the substrate. However, such tailored waveforms require complicated and expensive modification of current plasma systems. For example, rather than applying a simple waveform (such as a sine wave or a square wave) additional waveform generating circuitry must be included to generate more complicated waveforms for delivery to electrodes in the plasma system. Consequently, while some conventional systems and methods succeed to various extents in narrowing the IED at the substrate, they include different circuits and configurations than the embodiment plasma systems and methods described herein.

[0027] The embodiment systems and methods described herein may have various advantages over conventional systems and methods. For example, the embodiment systems and methods may be able to generate target ion energy with a desired narrow IED with minimum hardware updates (such as to existing etching systems, such as HARC etching systems) and a control method that is easy to implement. Another potential advantage of the embodiment systems and methods is the ability to adjust the shape of the plasma potential to match the shape of the substrate potential so that the substrate-plasma sheath voltage has a substantially flat profile (i.e., remains substantially constant). One example of where this may be particularly advantageous is the ion charging phases of a pulsed plasma process. That is, where desirable, the embodiment systems and methods may advantageously result in substantially constant sheath voltage during charging phase(s), providing narrower energy spread of ions traversing the plasma sheath.

[0028] Embodiments provided below describe various plasma systems, and in particular, to plasma systems that include a potential adjustment circuit coupled between a pulsed power supply and a plasma electrode. The following description describes the embodiments. FIG. 1 is used to describe an example plasma system. FIGS. 2A and 2B are used to describe several qualitative graphs for an example plasma system. Three more example plasma systems are described using FIGS. 3-5 while an example method of adjusting sheath potential of a plasma at a substrate is described using FIG. 6.

[0029] FIG. 1 illustrates an example plasma system including a plasma electrode configured to be

in contact with a plasma and a potential adjustment circuit coupled between the plasma electrode and a pulsed power supply in accordance with embodiments of the invention.

[0030] Referring to FIG. 1, a plasma system **100** includes a plasma electrode **110** (e.g., a top electrode) coupled to a pulsed plasma source power supply **120**. The pulsed plasma source power supply **120** is configured to deliver pulsed power (e.g., DC or RF) to the plasma electrode **110** through a potential adjustment circuit **130** coupled therebetween (i.e. along a transmission path **124** of the pulsed plasma power). The plasma system **100** also includes a substrate holder **114** configured to support a substrate **115**. The plasma system **100** is configured to deliver pulsed bias power **142** to the substrate **115** (such as by coupling the pulsed bias power **142** to the substrate holder **114**). The pulsed bias power **142** may be delivered using a pulsed bias power supply **140** (as shown), or may be delivered using the pulsed plasma source power supply **120**.

[0031] The plasma system **100** is configured to generate a plasma **112** between the plasma electrode **110** and the substrate **115** (whether by some combination of the pulsed plasma power and the pulsed bias power **142** or using other applied power). Both the plasma electrode **110** and the substrate **115** are in contact with the plasma **112**. The potential adjustment circuit **130** is configured to receive unadjusted pulsed plasma power **122** from the pulsed plasma source power supply **120** and adjust the electric potential profile (i.e. the shape of the voltage) of the pulsed power to generate adjusted pulsed plasma power **132** which is delivered to the plasma electrode **110**.

[0032] The plasma electrode **110** and the plasma **112** are electrically connected so that the plasma electrode **110** and the plasma **112** may have a certain substantially similar electric potential profile controlled by the pulsed plasma power and schematically represented as plasma potential **116**. Similarly, the substrate holder **114** is electrically connected to the substrate **115** and may have a certain substantially similar electric potential profile schematically represented as substrate potential **117** and that is controlled by the pulsed bias power **142**, but (as previously discussed) also influenced by other factors such as substrate material properties. A plasma sheath is formed at the interface between the plasma **112** and its boundaries (including the substrate **115**), schematically represented as sheath potential **118** and having an electric potential (i.e., sheath voltage) that is substantially equal to the difference between the plasma potential **116** and the substrate potential **117**.

[0033] The potential adjustment circuit **130** may include various passive elements, such as resistors, inductors, and capacitors. In various embodiments, the potential adjustment circuit **130** is an RLC circuit. The potential adjustment circuit **130** may include one or more of variable capacitance **134**, variable inductance **135**, and variable resistance **136**, such as variable components (i.e., dynamically adjustable, such as electronically adjustable, but may be mechanically adjustable). In various embodiments, the potential adjustment circuit **130** includes one or more variable capacitors, variable inductors, and variable resistors. A variety of combinations are possible. In one specific example, the potential adjustment circuit **130** has variable capacitance, inductance, and resistance (e.g., including at least one of each of a variable capacitor, variable inductor, and a variable resistor). In another specific example, the potential adjustment circuit **130** has variable capacitance, variable resistance, and fixed inductance, (e.g., at least one of each of a variable capacitor and a variable resistor, with one or more fixed inductors).

[0034] The electric potential profile of the substrate potential **117** may have a positive slope (i.e., potential vs. time, due to surface charging effects, for example). In contrast, the electric potential profile of the plasma potential **116** may have a substantially flat (or even negative) slope when the potential adjustment circuit **130** does not adjust the unadjusted pulsed plasma power **122** (i.e., if the unadjusted pulsed plasma power **122** is applied to the plasma electrode **110**). However, the potential adjustment circuit **130** may advantageously adjust the pulsed plasma power to generate the adjusted pulsed plasma power **132** so that the resulting slope of the plasma potential **116** is positive (e.g., matching the slope of the substrate potential **117** during the ion charging phases). This matched electric profile between the plasma potential **116** and the substrate potential **117** may

then produce a substantially flat slope for the sheath potential **118** resulting in a narrower IED. [0035] When included, the variable components of the potential adjustment circuit **130** may be varied within any suitable range of values. One example of a range of values that the variable capacitance **134** may be varied within is about 100 μF to about 1000 μF . An example of a range of values that the variable inductance **135** may be varied within is about 1 μH to about 1000 μH . Similarly, an example of a range of values that the variable resistance **136** may be varied within is about 1Ω to about 1000 Ω .

[0036] The pulsed power delivered by the plasma system **100** (e.g., using pulsed plasma source power supply **120** and pulsed bias power supply **140**) may be pulsed DC power or pulsed AC power, may be delivered in phase or out of phase, and may be pulsed at any desired frequency, such as an RF frequency. For example, the pulse frequency of the pulsed power may be on the order of kilohertz to hundreds of kilohertz (or higher). In various embodiments, the pulse frequency is on the order of hundreds of kilohertz, and is about 400 kHz in one embodiment. In various embodiments, the unadjusted pulsed plasma power **122** and the pulsed bias power **142** are supplied in phase at the same pulse frequency. One or both of the pulsed plasma source power supply **120** and the pulsed bias power supply **140** may include impedance matching networks (e.g., configured to efficiently coupled the pulsed power to the corresponding structures of the plasma system **100**).

[0037] The polarity of the pulses may also vary depending on a given application. In one embodiment, the pulsed bias power **142** is a series of negative DC pulses. However, in some cases, the pulsed bias power **142** may be or include positive DC pulses (such as to mitigate charging, for example). In one embodiment, the unadjusted pulsed plasma power **122** is a series of positive DC pulses. In another embodiment, the unadjusted pulsed plasma power **122** is a series of negative DC pulses. In still another embodiment, the unadjusted pulsed plasma power **122** is a series of alternating positive and negative DC pulses.

[0038] FIGS. 2A and 2B illustrate qualitative graphs of electric potential and IEDF for an example plasma system where FIG. 2A shows a broad IEDF for both low energy and high energy ions resulting from an unadjusted plasma potential profile and FIG. 2B shows a narrow IEDF for high energy ions resulting from an adjusted plasma potential profile in accordance with embodiments of the invention. The qualitative graphs of FIGS. 2A and 2B may be qualitatively representative of any of the example plasma systems described herein such as the plasma system of FIG. 1, for example. Similarly labeled elements may be as previously described.

[0039] Referring to FIGS. 2A and 2B, a qualitative potential profile graph **201** shows three time-resolved (time-dependent) voltage measurements for a first case where the pulsed power (here, pulsed DC power) applied to a plasma electrode is unadjusted. In this specific example, synchronized pulsed DC power (e.g., 400 kHz, at a 50% duty cycle) is applied to the bias electrode and the plasma electrode, respectively. During the substrate charging phases, a substrate potential profile **280** has a characteristic positive slope, shown as a substrate potential slope angle **290**, whereas an unadjusted plasma potential profile **281** has a flat slope, shown as an unadjusted plasma potential slope angle **291**, (which as previously mentioned may also be negative in some cases). The resulting shape of an unadjusted sheath potential profile **282** is a sloped sheath potential **292**, which causes a broad IED region **295** for high energy ions, as illustrated in the unadjusted IEDF **285** of qualitative IED graph **202**.

[0040] A qualitative potential profile graph **203** shows a second case where the pulsed power to the plasma electrode is adjusted. For this example, during the ion charging phases, the substrate potential profile **280** still has the positive slope, but adjusted plasma potential profile **283** has a positive slope (e.g., with an adjusted plasma potential slope angle **293** matching the substrate potential slope angle **290** of the substrate potential profile **280**, as shown). Now the resulting shape of adjusted sheath potential profile **284** is a substantially flat sheath potential profile **294**, which causes a narrow IED region **296** for high energy ions, as illustrated in the adjusted IEDF **286** of qualitative IED graph **204**.

[0041] As demonstrated, the slope of the substrate potential profile relative to the slope of the plasma potential profile (which may also be instantaneous in the case of less linear voltage response) may be used as indicators of the flatness of the sheath potential. For this reason, time-resolved measurements of the electric potential at the plasma electrode and the substrate (or substrate holder) may be used to inform the nature of the adjustment of the plasma potential (e.g., using a potential adjustment circuit).

[0042] FIG. 3 illustrates an example plasma system including a plasma electrode, a potential adjustment circuit, one or more sensors configured to obtain time-resolved electric potential information received by a controller configured to control the potential adjustment circuit in accordance with embodiments of the invention. The plasma system of FIG. 3 may be a specific implementation of other plasma systems described herein such as the plasma system of FIG. 1, for example. Similarly labeled elements may be as previously described.

[0043] Referring to FIG. 3, a plasma system **300** includes a plasma electrode **310** coupled to a pulsed DC source power supply **320** configured to deliver pulsed DC plasma power to the plasma electrode **310** through a potential adjustment circuit **330** coupled along a transmission path **324**. It should be noted that here and in the following a convention has been adopted for brevity and clarity wherein elements adhering to the pattern [x30] where 'x' is the figure number may be related implementations of a potential adjustment circuit in various embodiments. For example, the potential adjustment circuit **330** may be similar to the potential adjustment circuit **130** except as otherwise stated. An analogous convention has also been adopted for other elements as made clear by the use of similar terms in conjunction with the aforementioned numbering system.

[0044] The plasma system **300** also includes a substrate holder **314** configured to support a substrate **315** and a pulsed DC bias power supply **340** configured to deliver pulsed DC bias power **342** to the substrate **315** by coupling the pulsed DC bias power **342** to the substrate holder **314**. The plasma system **300** is configured to generate a plasma **312** between the plasma electrode **310** and the substrate **315** that is in contact with both the plasma electrode **310** and the substrate **315**. A controller **360** is configured to control the pulsed DC source power supply **320** to provide an unadjusted pulsed DC plasma power **322** (here, a series of positive DC pulses **323**, which are the electric potential profile of the unadjusted pulsed DC plasma power **322** prior to adjustment by the potential adjustment circuit **330**) to the potential adjustment circuit **330** and to control the pulsed DC bias power supply **340** to provide pulsed DC bias power **342** (here, a series of negative DC pulses **343**) to the substrate holder **314**.

[0045] One or more sensors **350** are also included and configured to measure the electric potential in the plasma system **300**. For example, a plasma potential sensor **352** (e.g. a high voltage probe) may be included to measure plasma potential **316**, a substrate potential sensor **354** (e.g. a high voltage probe or a voltage-current sensor) may be included to measure substrate potential **317** (e.g., at the substrate holder **314**), and so on. The sheath potential **318** may be measured indirectly (e.g., but subtracting the substrate potential **317** from the plasma potential **316**).

[0046] The sensors **350** are configured to obtain time-resolved electric potential information **356**, which is then provide to the controller **360**. Based on the time-resolved electric potential information **356**, the controller **360** is configured to control the potential adjustment circuit **330** (which includes various controllable elements allowing some combination of variable capacitance **334**, variable inductance **335**, and variable resistance **336**) to adjust the pulsed DC plasma power and generate adjusted DC pulsed plasma power **332**, which is delivered to the plasma electrode **310** along the transmission path **324**.

[0047] The sensors **350** may take additional measurements, such as monitoring the electric potential of the unadjusted pulsed DC plasma power **322**, the pulsed DC bias power **342**, etc. The sensors **350** may also include additional sensors configured to measure other parameters, such as current, plasma density, plasma temperature, ion energy, and others. The sensors **350** may also provide this information to the controller **360**. The measurements including the measurements of

the plasma potential **316** and the substrate potential **317** may be direct or indirect.

[0048] The controller **360** may be implemented as a single controller (as schematically shown) or as a control system including various controllers (such as a dedicated potential adjustment circuit controller, pulse controller, etc.) and processors (whether local and remote). The controller **360** may include a processor **362** and/or a memory **364** (i.e., a non-transitory computer-readable medium) that stores a program including instructions that, when executed by the processor **362**, control the potential adjustment circuit **330** (as well as other components of the plasma system **300**, such as the pulsed DC source power supply **320** and the pulsed DC bias power supply **340**). For example, the memory **364** may have volatile memory (e.g., random access memory (RAM)) and non-volatile memory (e.g., flash memory). Alternatively, the program may be stored in physical memory at a remote location, such as in cloud storage. The processor **362** may be any suitable processor, such as the processor of a microcontroller, a general-purpose processor (such as a central processing unit (CPU), a microprocessor, a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), and others.

[0049] The controller **360** may be further configured to control other parameters associated with performing a desired plasma process using the plasma system **300**, such as pulse frequency, duty cycle, pulse amplitude (e.g., peak voltage) for both the unadjusted pulsed DC plasma power **322** and the pulsed DC bias power **342**, for example. In one specific example, adjusting the electric profile of the pulsed DC plasma power may alter the peak ion energy (e.g., away from a target ion energy). The controller **360** may then be configured to adjust the pulsed power to one or more of the plasma electrode **310** and the substrate holder **314** to adjust the peak ion energy and bring it to the desired target ion energy (e.g., to within a predetermined tolerance. for example).

[0050] FIG. **4** illustrates an example capacitively coupled plasma (CCP) plasma system that includes a plasma electrode, a potential adjustment circuit, one or more sensors, and a controller in accordance with embodiments of the invention. The plasma system of FIG. **4** may be a specific implementation of other plasma systems described herein such as the plasma system of FIG. **3**, for example. Similarly labeled elements may be as previously described.

[0051] Referring to FIG. **4**, a CCP plasma system **400** includes a UEL **410** (an upper electrode that is a specific implementation of a plasma electrode) coupled to a pulsed DC source power supply **420** configured to deliver pulsed DC plasma power to the UEL **410** through a potential adjustment circuit **430**. The CCP plasma system **400** also includes an LEL **414** (a lower electrode that is a specific implementation of a substrate holder) configured to support a substrate **415** and a pulsed DC bias power supply **440** configured to deliver pulsed DC bias power **442** to the substrate **415** by coupling the pulsed DC bias power **442** to the LEL **414**. The CCP plasma system **400** is configured to generate a plasma **412** between the UEL **410** and the substrate **415** that is in contact with both the UEL **410** and the substrate **415**. The plasma **412** is contained by a plasma chamber **470**, which may be coupled to a ground potential **472**.

[0052] A controller **460** (which may include a processor **462** and memory **464**) is configured to control the pulsed DC source power supply **420** to provide an unadjusted pulsed DC source power **422** to the potential adjustment circuit **430** and to control the pulsed DC bias power supply **440** to provide pulsed DC bias power **442** to the LEL **414**. One or more sensors **450** are also included and configured to measure the electric potential in the CCP plasma system **400**, which may include a plasma potential sensor **452** and a substrate potential sensor **454**. The sensors **450** are configured to obtain time-resolved electric potential information **456**, which is then provide to the controller **460**. Based on the time-resolved electric potential information **456**, the controller **460** is configured to control the potential adjustment circuit **430** to adjust the pulsed DC source power and generate adjusted pulsed power **432**, which is delivered to the UEL **410**.

[0053] FIG. **5** illustrates an example inductively coupled plasma (ICP) together with a capacitively coupled plasma (CCP) system that includes a source power coupling element, a plasma electrode, a potential adjustment circuit, one or more sensors, and a controller in accordance with embodiments

of the invention. The plasma system of FIG. 5 may be a specific implementation of other plasma systems described herein such as the plasma system of FIG. 3, for example. Similarly labeled elements may be as previously described.

[0054] Referring to FIG. 5, an ICP/CCP plasma system **500** includes a plasma electrode **510** coupled to a pulsed DC source power supply **520** configured to deliver pulsed DC power to the plasma electrode **510** through a potential adjustment circuit **530**. The ICP/CCP plasma system **500** also includes a substrate holder **514** configured to support a substrate **515** and a pulsed DC bias power supply **540** configured to deliver pulsed DC bias power **542** to the substrate **515** by coupling the pulsed DC bias power **542** to the substrate holder **514**. The ICP/CCP plasma system **500** is configured to generate a plasma **512** between the plasma electrode **510** and the substrate **515** that is in contact with both the plasma electrode **510** and the substrate **515**. The plasma **512** is contained by a plasma chamber **570**, which may be coupled to a ground potential **572**.

[0055] A controller **560** (which may include a processor **562** and memory **564**) is configured to control the pulsed DC source power supply **520** to provide an unadjusted pulsed DC power **522** to the potential adjustment circuit **530** and to control the pulsed DC bias power supply **540** to provide pulsed DC bias power **542** to the substrate holder **514**. One or more sensors **550** are also included and configured to measure the electric potential in the ICP/CCP plasma system **500**, which may include a plasma potential sensor **552** and a substrate potential sensor **554**. The sensors **550** are configured to obtain time-resolved electric potential information **556**, which is then provide to the controller **560**. Based on the time-resolved electric potential information **556**, the controller **560** is configured to control the potential adjustment circuit **530** to adjust the pulsed DC power and generate adjusted DC pulsed power **532**, which is delivered to the plasma electrode **510**.

[0056] In contrast to the CCP plasma system **400**, the ICP/CCP plasma system **500** is configured to generate the plasma **512** using an alternative mechanism that inductively couples source power (e.g., AC power, such as RF power, from a source power supply **578**) to a source power coupling element **574** (e.g., a resonant structure, such as an RF antenna). Various configurations may exist to accomplish this (e.g., along with including a plasma electrode **510**). For example, the source power coupling element **574** may be implemented as a planar coil **575** that couples the source power to the plasma **512** through an intervening dielectric material **579** (e.g., a dielectric window). In other embodiments, the source power coupling element **574** may be implemented as a helical coil **576** (which also much be coupled through a dielectric material, such as a dielectric cylinder (not shown)).

[0057] FIG. 6 illustrates an example method of adjusting sheath potential of a plasma at a substrate in accordance with embodiments of the invention. The method of FIG. 6 may be performed by the example plasma systems described herein, such as the plasma system of FIG. 3, for example. The method of FIG. 6 may be combined with other methods and performed using the systems and apparatuses as described herein. For example, the method of FIG. 6 may be combined with any of the embodiments of FIGS. 1-5. Although shown in a logical order, the arrangement and numbering of the steps of FIG. 6 are not intended to be limited. The method steps of FIG. 6 may be performed in any suitable order or concurrently with one another as may be apparent to a person of skill in the art.

[0058] Referring to FIG. 6, a method **600** of adjusting sheath potential of a plasma at a substrate includes a pulsing step **601** of concurrently applying pulsed power to a plasma electrode in a plasma pulsing step **604** while applying pulsed power to the substrate (e.g., bias power) in a substrate pulsing step **605**. The plasma may be generated in a plasma chamber of a plasma system, which may be any type of plasma system, such as a CCP plasma system, an ICP plasma system, and others. The plasma electrode is in contact with the plasma and the substrate may be configured to be processed by the plasma (e.g., using various plasma-driven or plasma-enhanced processing techniques, such as etching, deposition, cleaning, modification, etc.). The pulsed source power and the pulsed bias power have the same pulse frequency in various embodiments. In one embodiment,

the pulsed power to the plasma electrode and the substrate are in phase, but the pulsed power may be delivered out-of-phase in other embodiments.

[0059] The applied power to the plasma electrode and the substrate may be independently selected to be pulsed DC (e.g. square waveform) or pulsed AC (e.g., sinusoidal waveform) power. In one embodiment, the pulsed power to both the plasma electrode and the substrate is pulsed DC power, but other configurations are possible. The pulsed power may be provided to the plasma electrode and to the substrate by a single power supply or by multiple power supplies. Further, one or more additional power supplies may also be included in the plasma system to supply power to other components. The pulsed bias power is a series of negative DC pulses in one embodiment. The pulsed plasma power may be any combination of positive and negative pulses. In one embodiment, the pulsed plasma power is a series of positive DC pulses. In another embodiment, the pulsed plasma power is a series of negative DC pulses. However, the pulsed plasma power may also alternate between positive and negative pulses in some embodiments.

[0060] In a measurement step **602**, the electric potential produced by the pulsed power (i.e., the pulsed plasma power and the pulsed bias power) is measured to obtain time-resolved electric potential information (e.g., using one or more sensors included in the plasma system). For example, one or more sensors may be configured to probe the electric potential (i.e., voltage) at various locations in of the plasma system, such as at the plasma electrode (with a plasma potential sensor) and the substrate (with a substrate potential sensor), to generate time-dependent potential signal(s) containing the time-resolved electric potential information. That is, as shown in FIG. **6**, the plasma potential profile as well as the substrate potential profile may be measured during the measurement step **602**. Other measurements (including additional potential measurements as well as measurements of other parameters of interest) may also be made during the measurement step **602** (and during other phases of the method **600**. In one embodiment measurements to estimate or determine the ion energy at the substrate are also made during the measurement step **602**.

[0061] The electric potential profile of the pulsed power to the plasma electrode (i.e., shape of the voltage applied at the plasma electrode) is then adjusted in a potential adjustment step **603** (using a potential adjustment circuit coupled between a pulsed plasma power supply and the plasma electrode along a transmission path of the pulsed plasma power, for example). For instance, the pulsed plasma power potential profile may be adjusted based on the time-resolved electric potential information obtained in the measurement step **602**. Adjusting the electric potential of the pulsed power to the plasma electrode results in an adjustment to the sheath potential of the plasma at the substrate. As already discussed in the foregoing, this may result in a substantially flat sheath potential profile, which may advantageously narrow the IED of ions at the substrate.

[0062] The potential adjustment circuit may include various electrical components (such as resistors, inductors, capacitors, and the like) that can be adjusted to adjust transmission properties of the transmission path for the pulsed plasma power. For example, the potential adjustment circuit may be or include an RLC circuit. In one embodiment, adjusting the electrical potential in the potential adjustment step **603** includes adjusting capacitance along the transmission path. In another embodiment, adjusting the electrical potential in the potential adjustment step **603** includes adjusting inductance along the transmission path while in still another embodiment, adjusting the electrical potential in the potential adjustment step **603** includes adjusting resistance along the transmission path. Of course, adjustment of one property is not mutually exclusive and adjustment of multiple properties (e.g., capacitance and resistance) may be leveraged to advantageously perform potential adjustments over a broad range of potential profiles and processing conditions. Some components of the potential adjustment circuit may be fixed. In one embodiment, the potential adjustment circuit includes fixed inductance.

[0063] It should also be mentioned that the electrical components of a potential adjustment circuit may be preset as an optional preliminary preset step **609**. The preset values may be initial values that serve as a starting point for the potential adjustment in the potential adjustment step **603**. In

one embodiment, the plasma system may store one or more previously optimized sets of parameters for the electrical components of the potential adjustment circuit, which can be used in the preset step **609** as a more optimal starting point (e.g., requiring little or no adjustment due to similarities in substrates or processes).

[0064] The potential adjustment step **603** may be performed continuously (e.g., at some predetermined frequency which may or may not be correlated with the frequency of potential measurements). Alternatively, the potential adjustment step **603** may be performed if the difference between the slope angle of the plasma potential and the substrate potential is greater than a predetermined tolerance. For example, the time-resolved electric potential information may include the plasma potential profile and the substrate potential profile. A plasma potential slope ($\theta_{\text{sub.plasma}}$) and a substrate potential slope ($\theta_{\text{sub.substrate}}$) may then be determined.

[0065] In a potential profile check **606**, the difference between the plasma potential slope and the substrate potential slope may be determined (e.g., dictating how the electric potential of the pulsed power to the plasma electrode should be adjusted). If the magnitude of the difference ($|\theta_{\text{sub.plasma}} - \theta_{\text{sub.substrate}}|$) is less than the tolerance, then the potential may not be adjusted, while the difference being greater or equal to the tolerance may result in adjustment in the potential adjustment step **603**. The difference between the slopes may be minimized (i.e., the difference may be decreased incrementally until no additional benefit is gained by adjusting the electric potential). For example, the predetermined tolerance may represent a boundary value for the difference between the slopes that indicates that the difference is minimized.

[0066] Similarly, whether a target ion energy has been reached may be determined in an ion energy check **607**. For example, the target ion energy may be any value and may depend on the details of a given application. One example range for the target ion energy is in the range of about 1 eV to about 10 keV, such as about 3 keV. As previously mentioned, the ion energy at the substrate may be measured (indirectly or directly, at some time during the method **600**, such as during the measurement step **602**, but also potentially at other times). If the measured ion energy is unequal to a target ion energy (e.g., not within a predetermined tolerance of the target ion energy), then the adjustment of the pulsed power to one or both of the plasma electrode and the substrate may be performed in a power adjustment step **608**. If no adjustment is needed, then the power adjustment step **608** may be omitted.

[0067] At this stage, the measurement and adjustment process (e.g., the measurement step **602** and the potential adjustment step **603** as well as the potential profile check **606**, the ion energy check **607**, and the power adjustment step **608**, when included) may be repeated as part of a cycle. For example, this may be used if fluctuations in the slopes of either the plasma potential or the substrate potential are likely to occur during a given process, whether through some aspect of the process itself (such as a change in the rate of charging of the substrate due to etching) or systemic variability in the plasma system. However, in other cases, the slope angles may remain constant or substantially constant throughout a given process and the adjusted settings of the electrical components of the potential adjustment circuit may not need to be adjusted once they are set.

[0068] The method **600** may be performed by one or more controllers operatively coupled to a potential adjustment circuit and at least one pulsed power supply. The controller may include a processor and a non-transitory computer-readable medium storing a program including instructions that, when executed by the processor, perform the method **600**. Alternatively, the processor may be configured to execute a program from a computer-readable medium that is not included in the controller, whether included in a general-purpose computer operatively connected to the controller, or in a remote location, such as in cloud storage.

[0069] Example embodiments of the invention are summarized here. Other embodiments can also be understood from the entirety of the specification as well as the claims filed herein.

[0070] Example 1. A method of adjusting sheath potential of a plasma at a substrate, the method including: applying pulsed plasma source power to a plasma electrode in contact with the plasma

while applying pulsed bias power to the substrate; measuring electric potential produced by the pulsed plasma source power and the pulsed bias power to obtain time-resolved electric potential information; and adjusting an electric potential profile at the plasma electrode based on the time-resolved electric potential information to adjust the sheath potential of the plasma at the substrate.

[0071] Example 2. The method of example 1, where adjusting the electric potential profile at the plasma electrode of the pulsed plasma source power results in a substantially flat sheath potential profile.

[0072] Example 3. The method of one of examples 1 and 2, where the time-resolved electric potential information includes a plasma potential profile and a substrate potential profile, and where adjusting the electric potential profile at the plasma electrode includes minimizing the difference between a slope of the plasma potential profile and a slope of the substrate potential profile.

[0073] Example 4. The method of one of examples 1 to 3, where the pulsed plasma power and the pulsed bias power are both pulsed direct current (DC) power.

[0074] Example 5. The method of example 4, where the pulsed plasma power includes a series of positive DC pulses.

[0075] Example 6. The method of one of examples 4 and 5, where the pulsed plasma power and the pulsed bias power have the same pulse frequency and are pulsed in phase with one another.

[0076] Example 7. The method of one of examples 1 to 6, where adjusting the electric potential profile includes adjusting capacitance along a transmission path of the pulsed plasma power.

[0077] Example 8. The method of one of examples 1 to 7, where adjusting the electric potential profile includes adjusting inductance along a transmission path of the pulsed plasma power.

[0078] Example 9. The method of one of examples 1 to 8, where adjusting the electric potential profile includes adjusting resistance along a transmission path of the pulsed plasma power.

[0079] Example 10. A plasma system including: a plasma electrode configured to be in contact with a plasma; a substrate holder configured to support a substrate in contact with the plasma; at least one pulsed direct current (DC) power supply configured to deliver pulsed plasma power to the plasma electrode and pulsed bias power to the substrate holder; and a potential adjustment circuit coupled between the plasma electrode and the at least one pulsed DC power supply, the potential adjustment circuit being configured to adjust an electric potential profile at the plasma electrode.

[0080] Example 11. The plasma system of example 10, further including: at least one sensor configured to measure electric potential produced by the pulsed plasma power and the pulsed bias power to obtain time-resolved electric potential information, the potential adjustment circuit being configured to adjust the electric potential profile based on the time-resolved electric potential information.

[0081] Example 12. The plasma system of example 11, where the time-resolved electric potential information includes a plasma potential profile and a substrate potential profile, where the at least one sensor includes a plasma potential sensor configured to measure the plasma potential profile and a substrate potential sensor configured to measure the substrate potential profile, and where the potential adjustment circuit is further configured to adjust the electric potential profile by minimizing the difference between a slope of the plasma potential profile and a slope of the substrate potential profile.

[0082] Example 13. The plasma system of one of examples 10 to 12, where the plasma system is a capacitively coupled plasma (CCP) plasma system, the plasma electrode being an upper electrode of the CCP plasma system.

[0083] Example 14. The plasma system of one of examples 10 to 13, further including: a source power coupling element configured to generate the plasma through an intervening dielectric material using applied source power, the plasma system being an inductively coupled plasma (ICP) plasma system.

[0084] Example 15. The plasma system of one of examples 10 to 14, where the potential

adjustment circuit is an RLC circuit including one or more of variable capacitance, variable inductance, and variable resistance.

[0085] Example 16. The plasma system of example 15, where the potential adjustment circuit includes variable capacitance, variable resistance, and fixed inductance.

[0086] Example 17. A plasma system including: a plasma electrode configured to be in contact with a plasma; a substrate holder configured to support a substrate in contact with the plasma; at least one pulsed power supply; a potential adjustment circuit coupled between the plasma electrode and the at least one pulsed power supply; at least one sensor; and a controller operatively coupled to the potential adjustment circuit and the at least one pulsed power supply, the controller including a processor and a non-transitory computer-readable medium storing a program including instructions that, when executed by the processor, perform a method including: using the at least one pulsed power supply, applying pulsed plasma power to the plasma electrode while applying pulsed bias power to the substrate; using the at least one sensor, measuring electric potential produced by the pulsed plasma power and the pulsed bias power to obtain time-resolved electric potential information; and using the potential adjustment circuit, adjusting an electric potential profile at the plasma electrode based on the time-resolved electric potential information to adjust a sheath potential of the plasma at the substrate.

[0087] Example 18. The plasma system of example 17, where adjusting the electric potential profile at the plasma electrode results in a substantially flat sheath potential profile.

[0088] Example 19. The plasma system of one of examples 17 and 18, where the time-resolved electric potential information includes a plasma potential profile and a substrate potential profile, and where adjusting the electric potential profile at the plasma electrode includes minimizing the difference between a slope of the plasma potential profile and a slope of the substrate potential profile.

[0089] Example 20. The plasma system of one of examples 17 to 19, where adjusting the electric potential profile includes adjusting one or more of capacitance, inductance, and resistance along a transmission path of the pulsed plasma power.

[0090] While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A method of adjusting sheath potential of a plasma at a substrate, the method comprising: applying pulsed plasma source power to a plasma electrode in contact with the plasma while applying pulsed bias power to the substrate; measuring electric potential produced by the pulsed plasma source power and the pulsed bias power to obtain time-resolved electric potential information; and adjusting an electric potential profile at the plasma electrode based on the time-resolved electric potential information to adjust the sheath potential of the plasma at the substrate.
2. The method of claim 1, wherein adjusting the electric potential profile at the plasma electrode of the pulsed plasma source power results in a substantially flat sheath potential profile.
3. The method of claim 1, wherein the time-resolved electric potential information includes a plasma potential profile and a substrate potential profile, and wherein adjusting the electric potential profile at the plasma electrode comprises minimizing the difference between a slope of the plasma potential profile and a slope of the substrate potential profile.
4. The method of claim 1, wherein the pulsed plasma power and the pulsed bias power are both pulsed direct current (DC) power.
5. The method of claim 4, wherein the pulsed plasma power comprises a series of positive DC

pulses.

6. The method of claim 4, wherein the pulsed plasma power and the pulsed bias power have the same pulse frequency and are pulsed in phase with one another.

7. The method of claim 1, wherein adjusting the electric potential profile comprises adjusting capacitance along a transmission path of the pulsed plasma power.

8. The method of claim 1, wherein adjusting the electric potential profile comprises adjusting inductance along a transmission path of the pulsed plasma power.

9. The method of claim 1, wherein adjusting the electric potential profile comprises adjusting resistance along a transmission path of the pulsed plasma power.

10. A plasma system comprising: a plasma electrode configured to be in contact with a plasma; a substrate holder configured to support a substrate in contact with the plasma; at least one pulsed direct current (DC) power supply configured to deliver pulsed plasma power to the plasma electrode and pulsed bias power to the substrate holder; and a potential adjustment circuit coupled between the plasma electrode and the at least one pulsed DC power supply, the potential adjustment circuit being configured to adjust an electric potential profile at the plasma electrode.

11. The plasma system of claim 10, further comprising: at least one sensor configured to measure electric potential produced by the pulsed plasma power and the pulsed bias power to obtain time-resolved electric potential information, the potential adjustment circuit being configured to adjust the electric potential profile based on the time-resolved electric potential information.

12. The plasma system of claim 11, wherein the time-resolved electric potential information comprises a plasma potential profile and a substrate potential profile, wherein the at least one sensor comprises a plasma potential sensor configured to measure the plasma potential profile and a substrate potential sensor configured to measure the substrate potential profile, and wherein the potential adjustment circuit is further configured to adjust the electric potential profile by minimizing the difference between a slope of the plasma potential profile and a slope of the substrate potential profile.

13. The plasma system of claim 10, wherein the plasma system is a capacitively coupled plasma (CCP) plasma system, the plasma electrode being an upper electrode of the CCP plasma system.

14. The plasma system of claim 10, further comprising: a source power coupling element configured to generate the plasma through an intervening dielectric material using applied source power, the plasma system being an inductively coupled plasma (ICP) plasma system.

15. The plasma system of claim 10, wherein the potential adjustment circuit is an RLC circuit comprising one or more of variable capacitance, variable inductance, and variable resistance.

16. The plasma system of claim 15, wherein the potential adjustment circuit comprises variable capacitance, variable resistance, and fixed inductance.

17. A plasma system comprising: a plasma electrode configured to be in contact with a plasma; a substrate holder configured to support a substrate in contact with the plasma; at least one pulsed power supply; a potential adjustment circuit coupled between the plasma electrode and the at least one pulsed power supply; at least one sensor; and a controller operatively coupled to the potential adjustment circuit and the at least one pulsed power supply, the controller comprising a processor and a non-transitory computer-readable medium storing a program including instructions that, when executed by the processor, perform a method comprising: using the at least one pulsed power supply, applying pulsed plasma power to the plasma electrode while applying pulsed bias power to the substrate; using the at least one sensor, measuring electric potential produced by the pulsed plasma power and the pulsed bias power to obtain time-resolved electric potential information; and using the potential adjustment circuit, adjusting an electric potential profile at the plasma electrode based on the time-resolved electric potential information to adjust a sheath potential of the plasma at the substrate.

18. The plasma system of claim 17, wherein adjusting the electric potential profile at the plasma electrode results in a substantially flat sheath potential profile.

- 19.** The plasma system of claim 17, wherein the time-resolved electric potential information includes a plasma potential profile and a substrate potential profile, and wherein adjusting the electric potential profile at the plasma electrode comprises minimizing the difference between a slope of the plasma potential profile and a slope of the substrate potential profile.
- 20.** The plasma system of claim 17, wherein adjusting the electric potential profile comprises adjusting one or more of capacitance, inductance, and resistance along a transmission path of the pulsed plasma power.
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